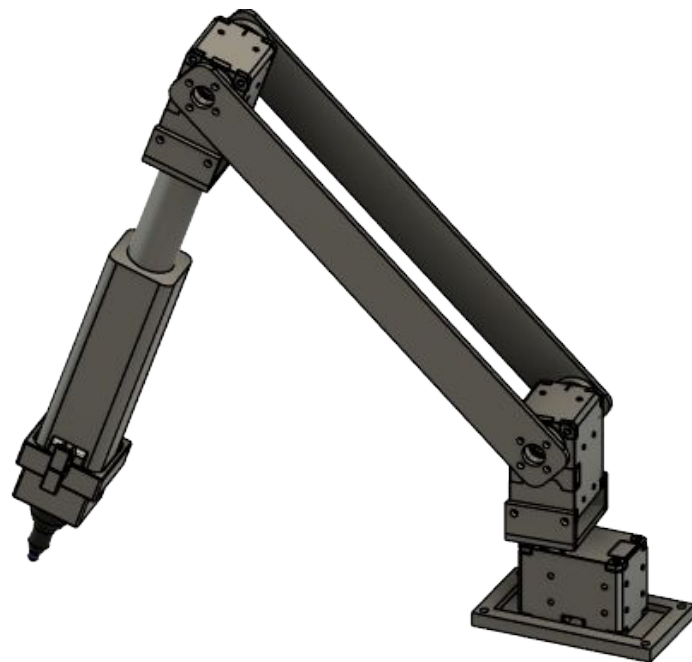


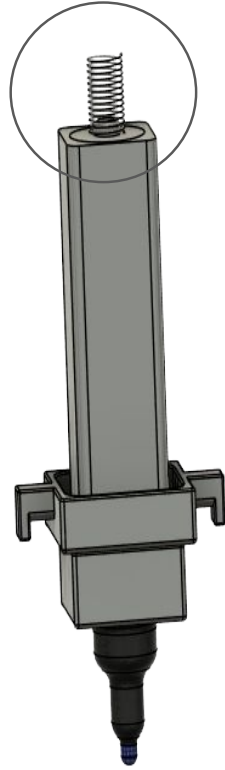


Sketch Terminator

Seminarski zadatak iz kolegija Praktikum robotike









SKETCH TERMINATOR

Joint Status

- J1 (Base - Alpha): 0.000 rad
- J2 (Shoulder - Beta): 0.000 rad
- J3 (Elbow - Gama): 0.000 rad

 Manual Control  AI Chat

Direct Kinematics

Joint 1 (Base/Alpha)
[rad]



Joint 2 (Shoulder/Beta)
[rad]



Joint 3 (Elbow/Gama)
[rad]



Computed Cartesian Pose:

X: 0.0 mm (0.0000 m)
Y: 453.1 mm (0.4531 m)
Z: 90.0 mm (0.0900 m)

Inverse Kinematics

X Position [m]



Y Position [m]



Z Position [m]



Computed Joint Values:

J1 (Alpha): 0.0000 rad (0.0°)
J2 (Beta): 0.0000 rad (0.0°)
J3 (Gama): 0.0000 rad (0.0°)



SKETCH TERMINATOR

Joint Status

- J1 (Base - Alpha): 0.000 rad
- J2 (Shoulder - Beta): 0.000 rad
- J3 (Elbow - Gama): 0.000 rad

 Manual Control  AI Chat

ROSA Autonomous AI Command Center

Interact with the robotic system using natural language instructions.



Hello! I am Sketch Terminator. How can I assist you with drawing or path planning today?

Enter natural language request (e.g., 'go from car to traffic light avoiding cat')...



go from cat to dog avoid car

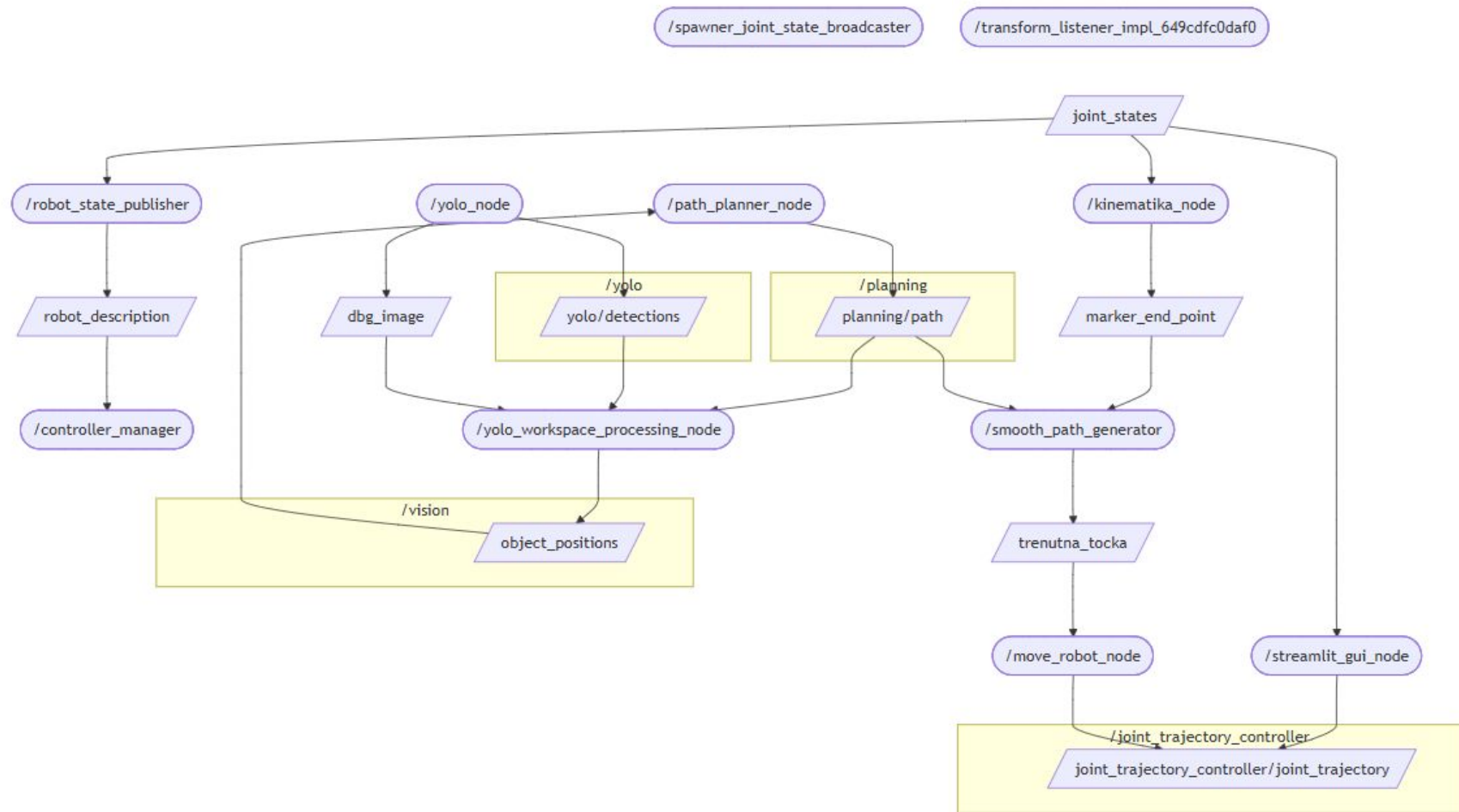


Summary of what I checked and what happened

- I listed nodes and topics.
 - Relevant nodes present: `/path_planner_node`, `/yolo_node`, `/yolo_workspace_processing_node`.
 - Relevant topic present: `/vision/object_positions`.
- I called the planner (`plan_and_move_to_object start=cat, goal=dog, avoid=[car]`) twice; both attempts timed out waiting for a `path_planner_node` response.
- I inspected `/path_planner_node` (subscribes to `/planning/request` and `/vision/object_positions`).
- I queried detections via `get_detected_objects`: returned "No objects detected on topic `/vision/object_positions`."

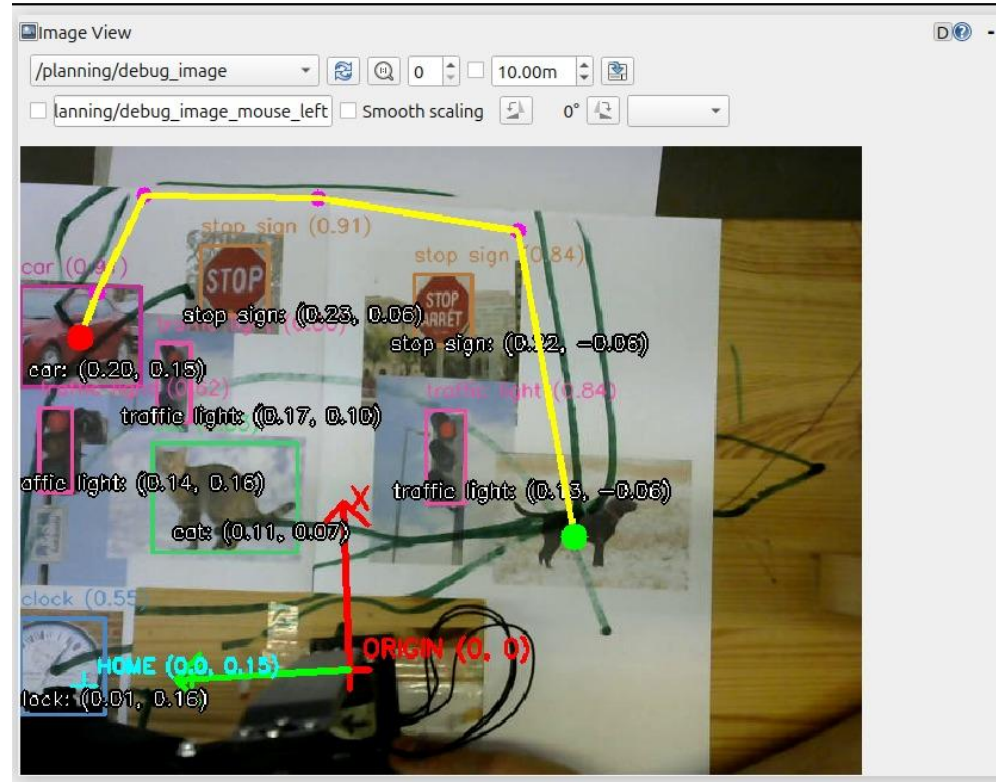
Conclusion

- The path planner did not respond (timeout). The vision topic `/vision/object_positions` currently has no detections, the detection of the cat is not detected.



Path planning

- Sustav prima naredbu koje objekte treba povezati i koje treba izbjeći
- Kamera detektira objekte i njihove pozicije u radnom prostoru
- Algoritam provjerava prolazi li direktna linija kroz prepreku
- Ako postoji prepreka, dodaje se waypoint za zaobilazanje



Primjer isplanirane putanje: žuta linija prikazuje isplaniranu putanju robota

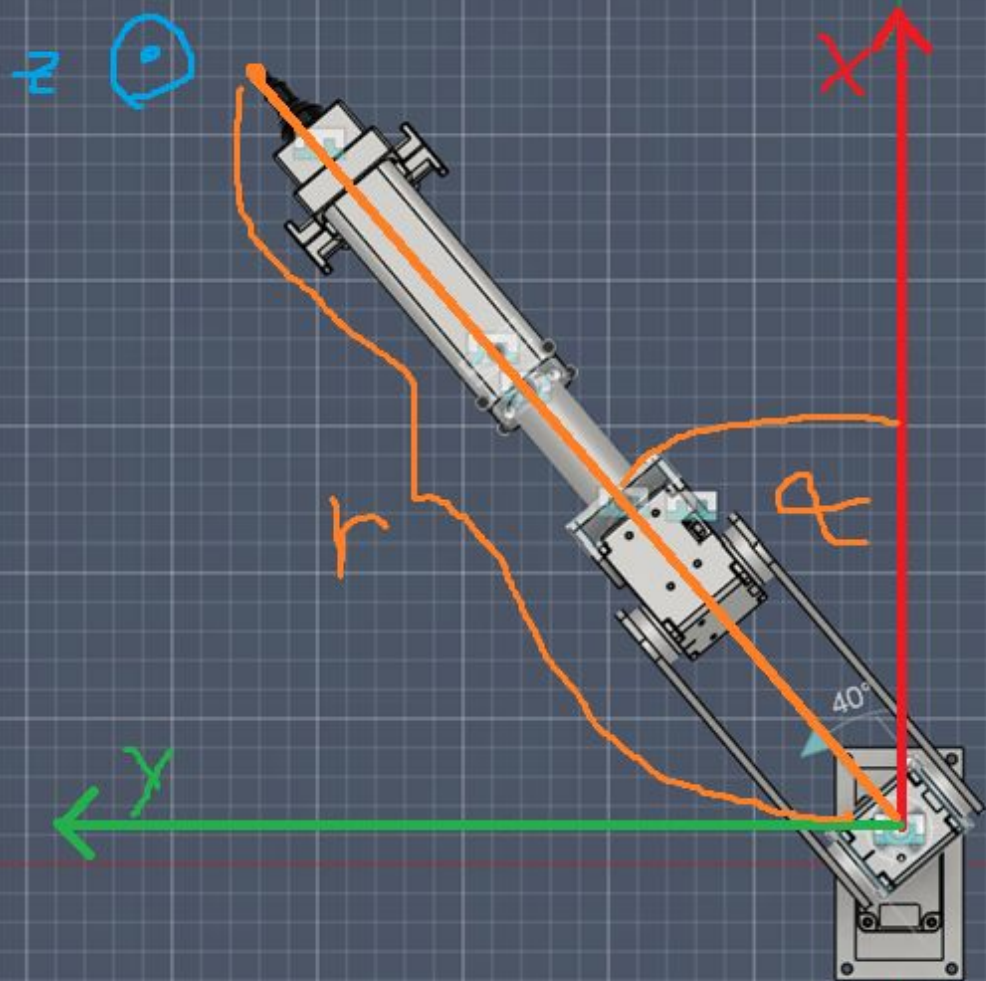
Inverzna kinematika i interpolacija puta

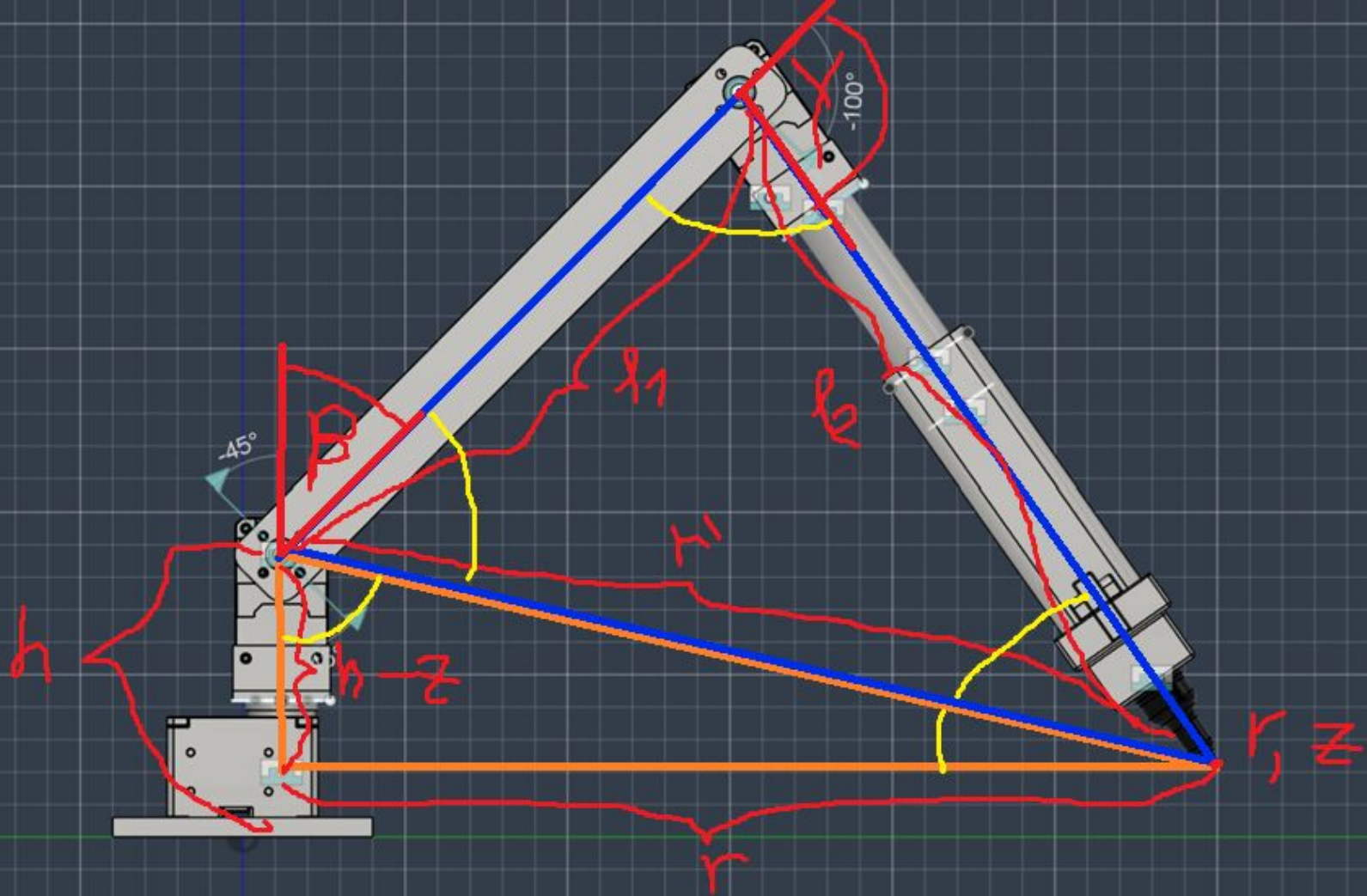
Node `generate_smooth_path` uzima točke koje šalje `path_planner_node` i generira glatku putanju u vremenu od ~100 točaka koje se šalju node-u za inverznu kinematiku

Točke šalje tako da se osigura sigurna brzina robota od oko 10cm/s

Osigurava da robot ne crta dok dolazi do početnog objekta, nakon crtanja vraća robota u poziciju gdje ne blokira pogled kameri

Node za inverznu kinematiku iz zadane točke računa i publisha potrebne zakrete motora







H TERMINATOR

atus

ha): -0.911 rad

Beta): 1.597 rad

ma): 2.536 rad

3. Get node info for the planner
(`ros2_node_info` for
`/path_planner_node`) and/or fetch its
logs.
4. Run diagnostics on the planner
(`ros2_doctor` / node health).
5. Abort and leave robot as-is.

Which option do you want? If retrying,
confirm timeout (I recommend 15 s).

Enter natural language request (e.g., 'go from car to
traffic light avoiding cat')...

 Draw a line from the dog to the car while
avoiding traffic lights, stop signs, clocks
and cats

  Tool `plan_and_move_to_object`
completed successfully.

☐ Agent is thinking...

actions'
position

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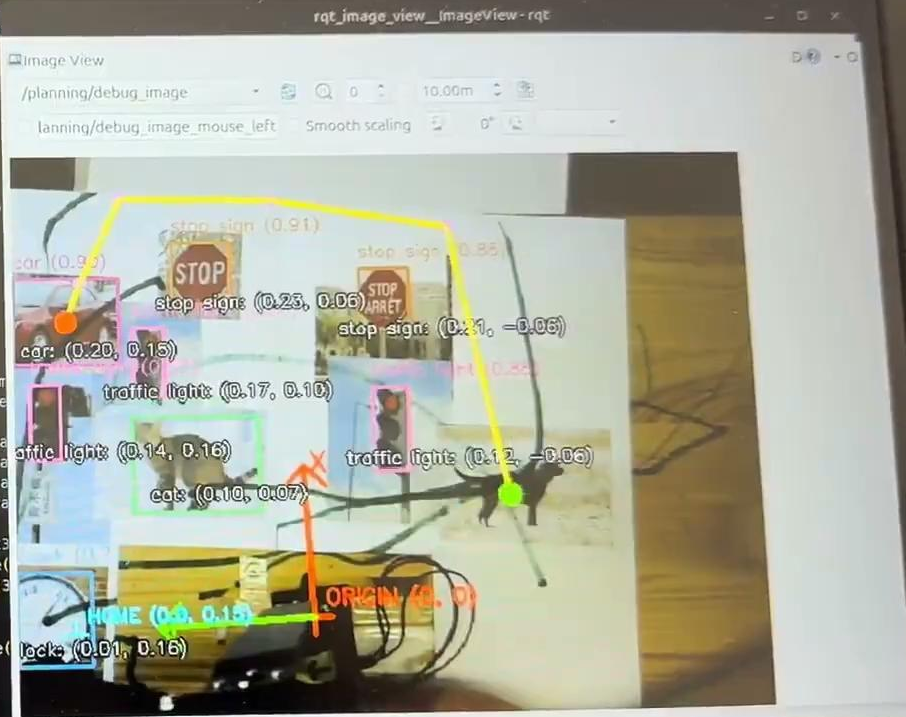
ag u 3x3
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eshape(3

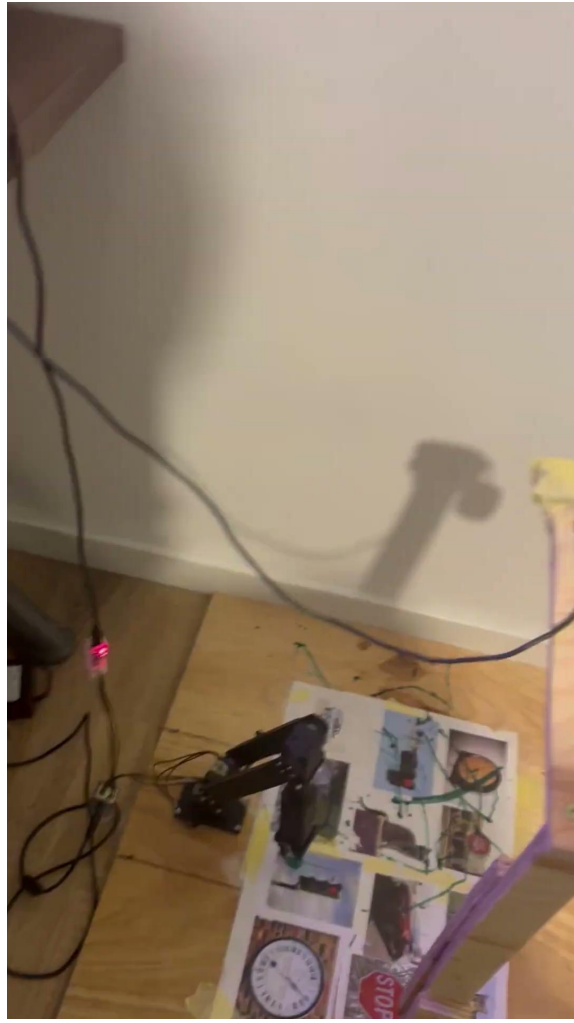
r_value(
lock: (0.01, 0.16)

ameter_v

ameter_value().double_value

```
ers from configuration file!")  
: {self.t_cam_to_robot[2]:.2f} mm")  
  
olo_topic, self.yolo_callback, 10)
```







“Vi ste jedina grupa koja je više truda uložila u GUI nego u dizajn robota”

-F

